WORKSHEET

Chapter 8: Normal distribution – Worded problems 2

Who first used the sign '=' to mean equals?

7	5	3	6	7	4	7	6	1	5	7	2	6

(Why did he use it? '… bicause noe 2 thynges can be moare equalle.')

Questions

1 Workers in a construction crew put together frames for a housing estate. They build a mean number of 40 frames per week with a standard deviation of 3.2.

a What is the probability that the crew will build more than 45 frames in a week?

b What is the probability that the crew will build between 34 and 38 frames in a week?

c They want to tell their foreman that 70% of the time they build at least x frames in a week. What is the value of x?

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2 A machine produces a mean of 112 defective parts in a day, with a standard deviation of 14.5.

a What is the probability that the machine produces fewer than 90 defective parts in a day?

b 85% of the time the machine produces fewer than how many defective parts?

3 The mean height of a class of 12-year-olds is 142 cm with a standard deviation of 4.8 cm.

a What is the probability that a student is taller than 135 cm?

b 25% of students are at least how many centimetres tall?

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Solutions

B 38

C 0.059

D 0.2356

E 0.9276

O 127

R 145

T 0.0647

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